

Workshop MaDiTraCe : machine learning applied to critical raw material traceability

23 — 24 April 2025

3 Av. Claude Guillemin, 45100 Orléans, France

Program

Wednesday, April 23rd

Thursday, April 24th

09:30

Hosting

10:00

Workshop Opening
SESSION I : post-analysis processing, feature creation and data completion

SESSION III : uncertainties propagation and Digital Product Passports

12:00

Workshop conclusion

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13:30

SESSION II : classification models

15:30

Labs. Visiting

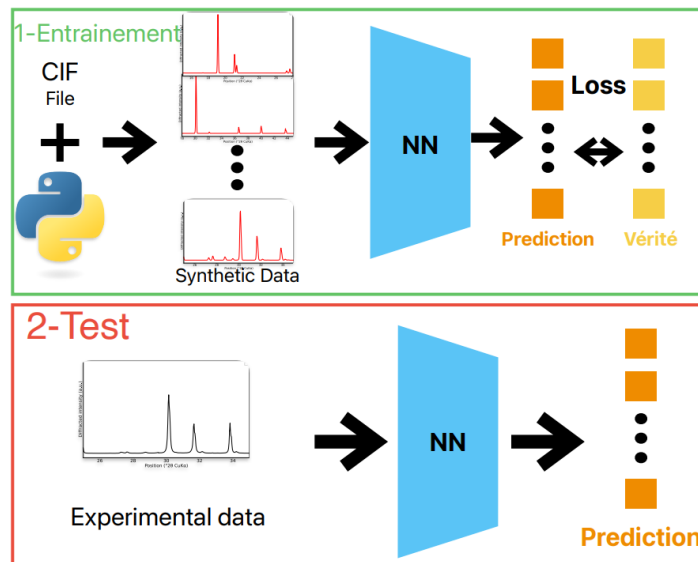
17:30

Session I : post-analysis processing, feature creation and data completion

Titouan Simonnet (BRGM) — **Neural networks for X-ray diffraction computed tomography. Application to Mineralogical Identification**

Understanding the chemical and mechanical behaviour of compacted materials (e.g. soil, subsoil, engineered materials) requires a quantitative description of the material's structure, and in particular the nature of the various mineralogical phases and their spatial relationships. Natural materials, however, are composed of numerous small-sized minerals, frequently mixed on a small scale.

Recent advances in synchrotron-based X-ray diffraction tomography now make it possible to obtain tomographic volumes with nanometer-sized voxels, with a XRD pattern for each of these voxels. Here, we use neural network approaches to identify and quantify minerals in a material. Training such models requires the construction of large-scale learning bases, which cannot be made up of experimental data alone. Algorithms capable of synthesizing XRD patterns to generate these bases have therefore been developed. An example of the analysis of an XRD-CT data set obtained on a concrete core will also be presented.



Herbert Rakotonirina (BRGM) — **Deep Internal Learning: Deep Learning with a Single Input**

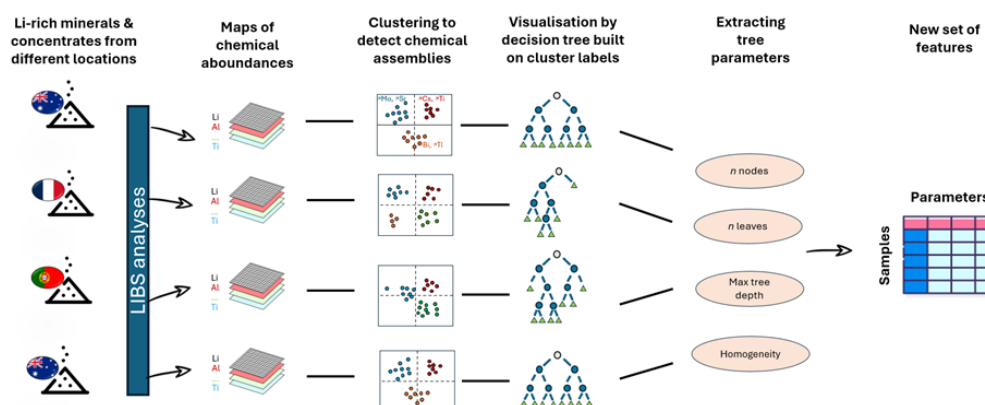
Deep Internal Learning is a paradigm where deep neural networks are trained on only a single input example, without relying on any external training data, by exploiting the input's own internal structure. This presentation introduces the concept and highlights three key methods that exemplify its power: (1) Deep Super-Resolution, a zero-shot approach that reconstructs a high-resolution image from one low-resolution input by leveraging internal patterns; (2) Deep Image Prior (DIP), which uses an untrained neural network's architecture as an implicit prior to perform image restoration tasks (such as denoising or inpainting) with no external data; and (3) a DIP-based spatial interpolation technique that infers missing spatial data solely from a given sample. Notably, none of these methods require any pre-collected training dataset, and the spatial interpolation approach is especially relevant in geoscience contexts where ground-truth observations are often scarce.

Nathan Bodereau (BRGM) — **Detecting chemical complexes from LIBS-based elemental imaging by coupling clustering and decision tree**

Compiling data from different devices could be challenging as it depends on their structures which could vary from a simply bulk data to massive and complex dataset. Here, we focus on the elemental images from Laser Induced Breakdown Spectroscopy (LIBS) performed on Li-rich minerals and concentrates. Combining high-resolution elemental images, that we seek to summarize, highlights elemental combinations related to sample specificities, such as minerals, linked to their provenance.

We developed an approach to detect these chemical complexes for each sample by coupling clustering and decision trees: The clustering allows identifying the main chemical complexes by grouping pixels by similarities while the decision tree, trained with cluster labels, is used to synthesize and standardize these complexes. Based on this work, we detected several elemental complexes that were not immediately obvious and constitute some indicators to classify samples according to their origins

How to summarize the main insights of geochemical maps ?



Session II: classification models

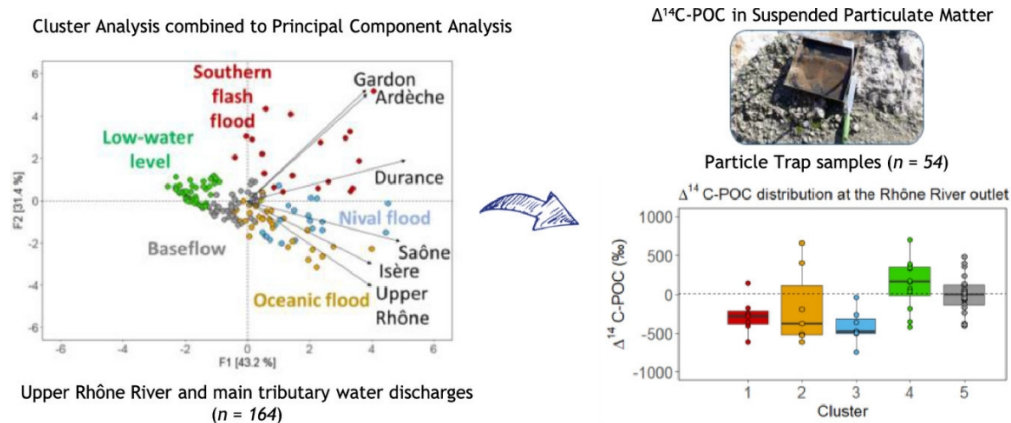
Alban Moradell-Casellas (BRGM) — **Lithium mines traceability with portable EDXRF: comparison of data exploitation by LDA, ACP, and RF**

A case study of the chemical data treatment from on-site instrument analysis (Portable EDXRF) for the lithium traceability will be presented. Two supervised models, Linear Discriminant Analysis (LDA) and Random Forest (RF), and an unsupervised model, Principal Component Analysis (PCA) were used to i) highlight the most discriminant features and ii) to distinguish samples according to their origins. Model comparisons are proposed, based on their performances, to better understand their respective contributions and limitations for the development of geochemical traceability systems. As a result, all these classification methods displayed similar configurations with several groups of samples related to their provenances. LDA et PCA however show limitations as they are sensitive to linear relationships between features and other relationships are not considered. Reversely, RF follows bimodal distributions that include any kind of relationship, and the risk of overfitting is avoided as samples are randomly selected to form sub-datasets to build the trees.

Nathan Bodereau (BRGM) — **A fuzzy clustering approach for sample classification: application to riverine radiocarbon in a nuclearized watershed.**

Fuzzy logic principle is a reliable method to cluster multidimensional data as it estimates contribution of each cluster on samples and so could be powerful, in contrast to traditional clustering where every sample is assigned to only one group. This approach has been applied to classify riverine particulate organic carbon-14 ($\Delta^{14}\text{C}$ -POC), at the outlet of the

Rhône River, linked to the origin of the material and nuclear power plant releases. Fuzz C-means Analysis was performed based on water discharges of the main tributaries. The classification summarized the river hydrology into five clusters with Mediterranean/Cevenol flood, oceanic pluvial flood, snowmelt flow, low-water level and baseflow clusters. $\Delta^{14}\text{C}$ -POC distribution is varied among each cluster confirming the impact of the hydrology on $\Delta^{14}\text{C}$ activity. In addition, the estimation of probabilities of sample belonging to each cluster made this methodology as a solution to estimate the contribution of each hydrology on $\Delta^{14}\text{C}$ -POC variation.



Thi Hai Yen Vu (BRGM, Orléans Univ.) — **Features Leverage in Graph Models for Mineral Prospectivity Mapping**

Mineral Prospectivity Mapping (MPM), the process of identifying areas with high potential for mineral deposits, can be divided into two main categories: knowledge-driven and data-driven. Knowledge driven techniques rely on expert opinion on geological data, while data-driven techniques employ ML models to predict the probabilities of mineral occurrences based on known geological datasets. Recently, with the advancement of machine learning (ML) methods, data-driven MPM has gained significant improvements. Notably, graph-based approaches overcome the disadvantages of previously used approaches (pixel-based, image-based) and have demonstrated better performances. However, the graph construction in current methods is based solely on spatial distances between pixels, regardless of their geological attributes. In this paper, we introduce a novel graph construction approach that combines spatial distances with other distances obtained from feature mining. Our experiments show that this combination outperforms existing graphs, and can be considered as a promising approach to integrate feature mining into data-driven models in MPM.

Session 3 : Uncertainties propagation and Digital Product Passports

Benjamin Le Teurnier (BRGM) — **Study of estimation uncertainty while using an autocalibration method for microgrid-based full Stokes imagers**

Full Stokes polarimetric images can be obtained from two acquisitions with a microgrid polarization camera equipped with a rotating retarder. When the delay of the retarder is imperfectly known, leading to an estimation bias, it can be calibrated from the measurements, but this requires three image acquisitions and may cause divergence of estimation variance at a low signal-to-noise ratio. With a study of the estimation uncertainties, we determine closed-form equations allowing one to decide in which experimental conditions this autocalibration is possible and useful, and to quantify the performance gain obtained in practice. These results are validated by simulations and real-world experiments.

Doruk Sahinel (Spherity) — **Building Trust for Artificial Fingerprinting with DPPs through Verifiable Credentials and AI Agent Authorization**

Digital Product Passports (DPP) build the digital backbone for the circular economy in Europe, enabling traceability for critical raw material supply chains. The MaDiTraCe project recognizes that DPPs can be more than just a regulatory tool, and ongoing research explores integrating material and artificial fingerprints into the DPP framework. While these fingerprints offer strong potential for anchoring physical products – and their geographical provenance – to their digital counterparts with high confidence, challenges remain in standardizing their representation. To certify artificial fingerprints, the project aims to create a verifiable metadata document describing the fingerprinting operation using W3C verifiable credentials. This includes details about the learning algorithms, model used, training data origin, and application area, stored as a credential in the DPP for auditability and transparency. Artificial fingerprint claims follow SSI principles and the concept will be enhanced by PoA credentials for AI agent authorization.